

HEMANT PACHAURI

Faridabad, India | +91 70423 26123 | a.hemant9002@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

BCA student (graduating 2026) at the intersection of full-stack engineering, applied AI/ML, and network security. Built an AI-driven intrusion detection system and a full-stack monitoring platform, backed by a 6-month full-stack internship and a co-authored research paper on AI in cybersecurity.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, C++, C#, PHP

Frontend: React, Vite, Tailwind CSS, HTML5, CSS3

Backend & APIs: Node.js, Express.js, Flask, REST APIs

Databases & ORMs: PostgreSQL, MySQL, MongoDB, SQLite, Prisma ORM

AI / ML: XGBoost, Scikit-learn, Isolation Forest, NumPy, Pandas

Cybersecurity & Infrastructure: Network Intrusion Detection, Packet Capture (Scapy), JWT/bcrypt Auth, pgbouncer Connection Pooling, Electron, Vercel, Render

PROJECTS

AI-POWERED CYBER THREAT ALERT SYSTEM

[GitHub Repository](#)

BCA Final Year Project — 3-member team — Supervisor: Dr. Shilpa Bhatia, MRIIRS

2025 – 2026

- Engineered a native Windows desktop intrusion detection system using Python Flask and Electron, capturing and analyzing live network packets in real time via Scapy
- Implemented a dual machine-learning pipeline — XGBoost trained on the CICIDS2017 dataset for DDoS, Port Scan, and Brute Force detection, paired with Isolation Forest for unsupervised zero-day anomaly detection
- Built an AI Explainability Engine that converts model predictions into human-readable threat reasoning, surfaced through native OS push notifications on detection
- Stack: Python, Flask, XGBoost, Scikit-learn, Scapy, Electron, SQLite, JavaScript, HTML5, CSS3

PULSEWATCH — UPTIME MONITORING PLATFORM

[GitHub Repository](#)

Personal Project — pnpm Monorepo

2026

- Shipped a full-stack uptime monitoring platform with a React/TypeScript frontend and a Node.js/Express backend, streaming live status updates to the dashboard via Server-Sent Events (SSE)
- Designed a normalized PostgreSQL schema (Supabase) using Prisma ORM, applying pgbouncer connection pooling to handle concurrent health-check writes from a node-cron scheduler
- Implemented JWT-based authentication with bcrypt password hashing and httpOnly refresh cookies, and enforced end-to-end request validation with Zod
- Stack: React, TypeScript, Vite, Tailwind CSS, Node.js, Express.js, PostgreSQL, Prisma, Zod — deployed on Vercel and Render

WORK EXPERIENCE

LABMENTIX PVT. LTD.

Bengaluru, Karnataka (Remote)

Full Stack Web Development Intern

June 2025 – December 2025

- Built 3+ responsive web applications using a React frontend and a Node.js/Express.js backend, shipping features across a 6-month development cycle
- Designed and maintained MySQL and MongoDB schemas, ensuring consistent backend-to-frontend data synchronization across modules
- Owned features end-to-end — UI design, implementation, testing, and deployment — completing all assigned modules within deadlines
- Recognised by the Founder & CEO for technical competence and professional dedication (Letter of Recommendation issued)

EDUCATION

MANAV RACHNA INTERNATIONAL INSTITUTE OF RESEARCH & STUDIES

Faridabad, India

Bachelor of Computer Applications (BCA)

2026 (Expected)

- Key subjects: Data Structures, OOP (C++, Java, C#), Web Development, AI Fundamentals, DBMS, Software Engineering, OS, IoT

ARAVALI INTERNATIONAL SCHOOL FARIDABAD

Faridabad, India

Senior Secondary (12th Grade)

2022

RESEARCH & CERTIFICATIONS

THREATS AND PREVENTION USING AI AND MACHINE LEARNING

2025

Co-authored Research Paper — Manav Rachna International Institute of Research & Studies — [Under Review]

- Reviewed supervised (SVM, Random Forest), deep learning (CNN, RNN/LSTM), and unsupervised models across the CICIDS2017 and UNSW-NB15 benchmark datasets for intrusion, phishing, and ransomware detection

Certifications: Introduction to Web Design & Development — Crafting Meaningful HTML — Programming Foundations: Fundamentals — Privacy & Security in Online Social Media (NPTEL)